

Assignment nr. 2 – T-junction

Assigned: Wednesday March 18th, 2026
Due: Wednesday April 1th, 2026 at class starting

Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to prepare **individually** the report of the lab, that must include the Matlab script, the obtained results and their brief comment;
- to submit the report **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal.

Problem nr. 1

Design a T-junction using ideal transmission lines with a characteristic impedance equal to $Z_{\infty 2}$ and $Z_{\infty 3}$ ($Z_{\infty 1} = Z_{r1}$) and with length equal to $\ell = \lambda/4$ at the frequency $f_0 = 5.8$ GHz. The specs are:

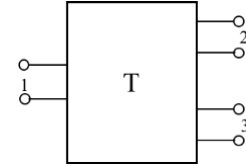
- port 1 matched ($S_{11} = 0$);
- output power ratio $P_3/P_2 = a$ with $a = 1.25$.

Assume the following reference impedances for the three ports:

$$Z_{r1} = Z_{r2} = Z_{r3} = Z_0 = 50 \Omega.$$

Draw the layout of the microstrip circuit (including the access line) using a dielectric material with $\epsilon_r = 4.45$ and $h = 1.55$ mm.

Using Matlab™, plot the frequency behavior of $|S_{11}(f)|$ as a function of f/f_0 in the range $[0, 2f_0]$.

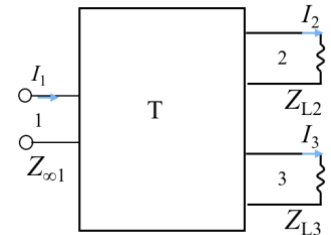


Problem nr. 2

Design a T-junction in such a way that, when the ports 2 and 3 are closed on the load Z_{L2} and Z_{L3} respectively, the following specs at the design frequency $f_0 = 5.8$ GHz are satisfied:

- port 1 matched ($S_{11} = 0$);
- $|I_3| = 1.25|I_2|$;
- $\angle I_3 = \angle I_2 - \pi$.

Assume reference impedance at port1 $Z_{\infty 1} = 50 \Omega$, while $Z_{L2} = 112.5 \Omega$ and $Z_{L3} = 90 \Omega$.



1. Select a configuration for the T-junction (with or without QWT).
2. Determine the characteristic impedance and the length of each line section necessary to design the selected T-junction configuration.
3. Draw the transmission line equivalent model of the T-junction.

4. Using Matlab™, plot the frequency behavior of the amplitude and phase of the ratio I_3/I_2 as a function of f/f_0 , in the range $[0, 2f_0]$.